Parameter-Estimation

By
Dr. Anjula Mehto
Assistant Professor
Thapar Institute of Engineering and Technology,
Patiala, Punjab
Email-anjula.mehto@thapar.edu

Parameter Estimation

- **What is unknown? → Parameters**
  (e.g., mean μ, variance σ^2, rate λ)
- **What do we have? → Sample data**
  (observations collected from experiments or real-world processes)
- **Goal → Estimate parameters from data**
  (using statistical estimation techniques)

Parameter Estimation

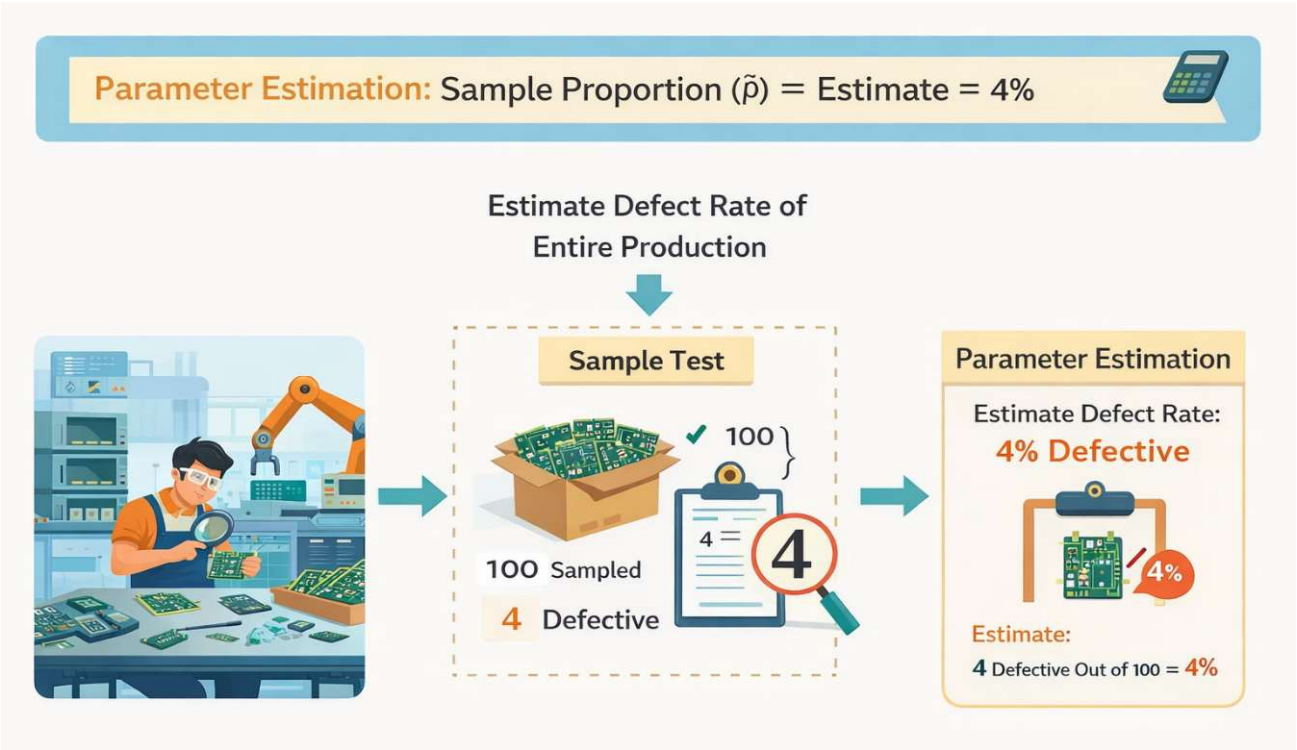
- Parameter Estimation is **a branch of statistics that involves using sample data to estimate the parameters of a distribution**.

Or

- Parameter estimation means using sample data to estimate unknown population values like mean and variance.
- **Parameter vs Statistic**
  - $\mu \rightarrow$ unknown (parameter)
  - $\bar{x} \rightarrow$ known (statistic)

Parameter Estimation

- *For instance, if a factory produces thousands of electronic components, instead of testing each item, quality control teams might randomly sample a certain number of items (e.g., 100 components) and check how many are defective.*
- *If they find that 4 out of 100 components are defective, they can estimate the proportion of defective items in the entire production batch as 4%.*



Parameters of Different Distributions

- The parameters are the numbers that yield the actual distribution.

Distribution	Parameters
Bernoulli(p)	$\theta = p$
Poisson(λ)	$\theta = \lambda$
Uniform (a,b)	$\theta = [a,b]$
Normal (μ, σ^2)	$\theta = [\mu, \sigma^2]$

Parameter Estimations Techniques

- Method of Moments (MoM)
- Maximum Likelihood Estimation (MLE)
- Minimum Mean Square Error (MMSE) Estimation

Moment

- A moment is a value calculated from data that tells us something about the distribution (like average, spread, etc.)

- **First Moment → Mean**
- Represents the **average value**
- Formula:

$$E(X)$$

- **Second Moment → Related to Variance**
- Represents **spread of data**
- Formula:

$$E\left(X^2\right)$$

- Compute variance:

$$Var(X) = E\left(X^2\right) - \left(E(X)\right)^2$$

Sample moments

- Calculated **directly from the observed data**
- They are **numbers we can compute**
- Examples:
- Sample mean:

$$\bar{x} = \frac{1}{n}\sum x_i$$

- Sample second moment:

$$\frac{1}{n}\sum x_i^2$$

- Sample mean: average of data
- Second moment: related to spread and used in variance calculation

Population moments

Population moments

- Defined **theoretically** using the probability distribution
- Written **in terms of unknown parameters**
- Examples:
- For a normal distribution:
  - Population mean = μ
  - Population variance = σ^2

Method of Moments

- **Estimating the Moments about population mean (μ).**
- Involves equating the sample moments of a statistical distribution to the corresponding population moments.
- The sample moments are computed from the observed data, while the population moments are expressed in terms of the parameters of the distribution.

Method of Moments

- Steps
  1. Find sample moments
  2. Write population moments
  3. Equate both
  4. Solve for parameters

- Match sample moments with population moments
- Sample *Moment* $=$ *Population Moment*

Method of Moments

- Suppose that we take a random sample from a rectangular distribution, i.e., a uniform distribution over [a, b]. The 2 parameters are ‘a’ and ‘b’.
- We have to estimate these 2 parameters given a sample of size ‘n’.

Method of Moments

- Let $(X_1, X_2, \ldots, X_n)$be a random sample of size '*n' taken from the variable 'X'.*
- E(*X) = (a + b)/2 =* $(x_1, + x_2 + \cdots + x_n)$*/n= m (say);*
- This gives us **a + b = 2m……(i)**

Take the average of the squares, then subtract the square of the average — that gives you variance."

- **Var(X)=E(X^2)-(E(X))^2**
- E(X^2)= Var(X) + (E(X))^2
- = (b-a)^2/12 + ((a + b)/2)^2
- = (a^2 + b^2 + ab)/3
- =(a^2 + b^2 + ab)/3 = p (say)
- **This gives us (a^2 + b^2 + ab) = 3p……(ii)**

Method of Moments

- Solving (i) and (ii) for a and b gives the estimates using method of moments.
- a = m -sqrt(3.p -3.m.m)
- b = 2.m –a = m + sqrt(3.p – 3.m.m)

- Or you can also write:

$$a = m - \sqrt{(3p - 3m^2)}$$
$$b = m + \sqrt{(3p - 3m^2)}$$

Method of Moments

- Let $(X1, X2, \ldots, Xn)$be a random sample of size '*n*' *taken from a Normal Population with parameters: mean* $\theta 1$ *and variance* $\theta 2$*.*
- Estimate these 2 parameters given a sample of size n ?

Method of Moments

- For the normal distribution, the population mean and variance are given by:
- $\mu = \theta 1$
- $\sigma^2 = \theta 2$
- The sample mean and variance can be calculated from the sample:
- $\bar{x} = (x_1 + x_2 + \ldots + x_n) / n$
- $s^2 = ((x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \ldots + (x_n - \bar{x})^2) / (n - 1)$

Method of Moments

- Setting the sample mean equal to the population mean, we get:
- $\bar{x} = \theta 1$
- Solving for $\theta 1$, we get:
- $\theta 1 = \bar{x}$
- Setting the sample variance equal to the population variance, we get:
- s^2 = $\theta 2$

Method of Moments

- Solving for $\theta 2$, we get:
- $\theta 2$ = s^2
- Therefore, the method of moments estimators for $\theta 1$ and $\theta 2$ are:
- $\theta 1$ = x̄ = (x_1 + x_2 + ... + x_n)/n (sample mean)
- $\theta 2$ = S^2 = ((x_1 - x̄)^2 + (x_2 - x̄)^2 + ... + (x_n - x̄)^2) / (n - 1) (sample variance)

Method of Moments

- **Method of Moments – Advantages-**
  - Simple
  - Quick estimates
- **Limitation**
  - Not always efficient

Maximum Likelihood Estimation

- The maximum likelihood estimation (MLE) estimates the population parameters by finding the values that maximize the likelihood function.
- For a discrete distribution, maximize the PMF of the data.
- For a continuous distribution, maximize the PDF of the data.

Maximum Likelihood Estimation

- Let us take a sample of size *'n' from an exponential population having the parameter 'λ'. So, 'X' is exponentially distributed with parameter 'λ'.*

Maximum Likelihood Estimation

- Solution-
- A sample of size *'n' is taken:* **($X_1, X_2, \ldots, X_n$)**
- Exponential density is:

$$f(x) = \lambda . e^{\wedge}(-\lambda . x), \quad \lambda > 0, x \geq 0$$

- The density of X_i is :

$$\lambda . e^{\wedge}(-\lambda . x_i), \quad i = 1, 2, \ldots . n.$$

- Joint density of ($X_1, X_2, \ldots, X_n$)
- **L(x_1, x_2, …., x_n)=** $\lambda \prod_{i=1}^{n} e^{\wedge}(-\lambda . x_i)$

Maximum Likelihood Estimation

- Take natural logarithm.
- ln(L(x1, x2, …., xn)) = ln(λ^n* e^((−$\boldsymbol{\lambda}.\boldsymbol{x}_\mathbf{1}$) + (−$\boldsymbol{\lambda}.\boldsymbol{x}_\mathbf{2}$) + ⋯ + (−$\boldsymbol{\lambda}.\boldsymbol{x}_\boldsymbol{n}$)))
- = n*ln($\boldsymbol{\lambda}$)+ ((−$\boldsymbol{\lambda}.\boldsymbol{x}_\mathbf{1}$) + (−$\boldsymbol{\lambda}.\boldsymbol{x}_\mathbf{2}$) +⋯ +(−$\boldsymbol{\lambda}.\boldsymbol{x}_\boldsymbol{n}$))
- = n*ln($\boldsymbol{\lambda}$)−$\boldsymbol{\lambda}$ ($\boldsymbol{x}_\mathbf{1}$+ $\boldsymbol{x}_\mathbf{2}$+..+$\boldsymbol{x}_\boldsymbol{n}$)

- **Differentiate this equation with respect to $\boldsymbol{\lambda}$, and equate to zero, this will give us the estimator that is obtained using this method of maximum likelihood estimation.**

Maximum Likelihood Estimation

- Suppose

  Z= n*ln($\boldsymbol{\lambda}$)–$\boldsymbol{\lambda}$ ($\boldsymbol{x_1}$+ $\boldsymbol{x_2}$+..+$\boldsymbol{x_n}$)

$$\frac{\partial Z}{\partial \lambda} = 0$$

  n/ $\boldsymbol{\lambda}$ - $\boldsymbol{\lambda}$ ($\boldsymbol{x_1}$+ $\boldsymbol{x_2}$+..+$\boldsymbol{x_n}$) = 0

  Take ($\boldsymbol{x_1}$+ $\boldsymbol{x_2}$+..+$\boldsymbol{x_n}$) = $\sum_{i=1}^{n} x_i$

  $\boldsymbol{\lambda}$=$\boldsymbol{1/sample(mean)}$

Maximum Likelihood Estimation

Q.1 Assume x_1, x_2, x_3……x_n be independent and identically distributed sample from a distribution with probability defined as:

P_(x_i) (x;θ)=(1-θ)^(x-1).θ, $= (1-\theta)^{(x-1)} \cdot \theta$

where θ is the parameter of the distribution. Find the estimator of the parameter using Maximum Likelihood Estimation.

Maximum Likelihood Estimation

Answer-

$$\theta_MLE = n / \sum(x_i)$$

$$\theta_MLE = 1/mean$$

Maximum Likelihood Estimation

- To find the maximum likelihood estimator of the parameter θ, we need to maximize the likelihood function, which is given by:
- $L(x_1, x_2, ..., x_n) = \prod[P_{(X_i)}(x_i;\theta)]$
- where $x_1, x_2, ..., x_n$ are the observed values of the random variables $X_1, X_2, ..., X_n$.
- Taking the natural logarithm of the likelihood function, we get:
- $\ln L(x_1, x_2, ..., x_n) = \sum[\ln P_{(X_i)}]$
- $= \ln (1-\theta)^{\sum(x_i-n)}.\theta^n$
- $=(\sum(x_i - n)) \ln(1 - \theta)]+ n. \ln(\theta)$

Maximum Likelihood Estimation

- To find the maximum of this function, we take the derivative with respect to θ and set it equal to zero:
- $d/d\theta\ [\ln L(x_1, x_2, ..., x_n)]$
- $= d/d\theta\ (\sum(x_i - n))\ \ln(1 - \theta) + n.\ \ln(\theta)]$
- $= (\sum(x_i - n))/(1 - \theta).(-1) + n/\theta] = 0$
- Simplifying this expression, we get:
- $(\sum(x_i - n))/(1 - \theta)] = n/\theta$
- Solving for θ, we get the maximum likelihood estimator:
- **$\theta_MLE = n / \sum(x_i)$**

Maximum Likelihood (ML) Estimation

Assignment Question-1

- Let $(X1, X2, \ldots, Xn)$be a random sample of size *n taken from a Normal Population with parameters: mean* $\theta 1$ *and variance* $\theta 2$*. Find the Maximum Likelihood Estimates of these two parameters.*

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{\frac{-(x-\mu)^2}{2\sigma^2}}$$

Maximum Likelihood Estimation

Assignment Question-2

- Let $X_1, X_2 \ldots, X_n$ be a random sample from $B(m, \theta)$ distribution, where $\theta \in \Theta = (0, 1)$ is unknown and 'm' is a known positive integer. Compute value of θ using the M.L.E.

Maximum Likelihood Estimation

- **MLE – Why popular**
  - Widely used in ML, signal processing

Minimum Mean Square Error (MMSE) Estimation

- **Why MMSE?**
- **MoM and MLE Focus on PARAMETERS**
- Estimate unknown parameters (μ, λ, etc.)
- Based on:
  - MoM → matching moments
  - MLE → maximizing likelihood

- MMSE is needed because it directly minimizes estimation error, whereas MoM and MLE focus on parameter estimation without explicitly minimizing error.

MoM vs MLE vs MMSE

Method	Idea	Complexity	Goal
MoM	Match moments	Easy	Solve equations
MLE	Maximize likelihood	Moderate	Fit data best
MMSE	Minimize error	Advanced	Minimize estimation error

Minimum Mean Square Error (MMSE) Estimation

- MMSE estimation finds the estimator that minimizes the expected value of the squared error between the true value and the estimate.

- Why do we square error?
- To avoid negative cancellation
- To penalize large errors more

- MMSE is a method used to estimate an unknown value such that the **average squared error is minimized**.
  - *Minimize* $E\left[\left(X-\hat{X}\right)^2\right]$

Minimum Mean Square Error (MMSE) Estimation

- Given some information that is related to an unknown quantity of interest (say random variable X) , the problem is to obtain a good estimate for the unknown (variable X) in terms of the observed data (say random variable Y)
- In general, our estimate **x'** is a function of y:

$$x' = g(y)$$

Minimum Mean Square Error Estimation

- The error ($\tilde{E}$) in our estimate is given by

$$\tilde{E} = X - x' = X - g(y)$$

- In MMSE the objective is to minimize the expected value of residual square, where residual is the difference between the true value and the estimated value.
- The expected residual square is also known as MSE (Mean Square Error):

$$E[(X-x')^2 | Y=y] = E[(X-g(y))^2 | Y=y]$$

Procedure to Solve MMSE

1. Define the estimator
2. Construct the MSE (Mean Square Error)
3. Differentiate the MSE with respect to the parameter and set it to zero.
4. Put the MMSE estimator in step 3 in the MSE expression to find the minimum residual square.

Minimum Mean Square Error Estimation

- For simplicity, let us first consider the case that we would like to estimate X without observing anything.
- What would be our best estimate of X in that case?
- Let **`a`** be our estimate of X. (Step-1)
- Then, the MSE is given by: (Step-2)

$$h(a) = E[(X-a)^2]$$

$$= E[X^2] + a^2 - 2.a.E[X] \quad \ldots Eq(1)$$

Minimum Mean Square Error Estimation

- This is a quadratic function of `**a**`, and we can find the minimizing value of `**a**` by differentiation and set it to zero (Step-3)

$$h'(a) = -2E[X] + 2a = 0$$

- Therefore, we conclude the minimizing value of `**a**` is:

$$a = E[X]$$

- Put the value of `a` in the Equation (1) (Step-4)

Minimum Mean Square Error Estimation

$$h(a) = E[X^2] + E[X]^2 - 2.E[X].E[X]$$
$$= E[X^2] + E[X]^2 - 2\,E[X]^2$$
$$= E[X^2] - E[X]^2$$
$$= Var(X)$$

- **Interpretation:** The best constant estimator of X is the expected value of X. The minimum MSE using the optimal estimator is the variance of X.

Hypothesis Testing

Hypothesis Testing

- Definition- "**A hypothesis may be simply defined as a guess. A scientific hypothesis is an intelligent guess**"
- **A statistical hypothesis is an assumption about a population parameter.**
- Hypothesis testing is a statistical method used to determine whether a hypothesis about a population parameter is supported by the available sample data.
- **This assumption about the parameter may or may not be true.**

Hypothesis Testing

- An example of hypothesis testing is testing whether the mean weight of apples in a particular fruit garden is 100 grams.

Types of Hypothesis

- In hypothesis testing, we set up two competing hypotheses:
  - **Null Hypothesis (H_0)**:
  - **Alternative Hypothesis (H_1 or H_a):**

Null Hypothesis

- **Null Hypothesis-**
- It is a statement that there is no significant difference or relationship between two populations or variables.
- **The null hypothesis, denoted by H_0.**
- Example- The null hypothesis is that the mean weight is 100 grams.

Alternative Hypothesis

- It is a statement that there is a significant difference or relationship between the two populations or variables.
- **The alternative hypothesis, denoted by H_1or H_a.**
- **There are three types of alternative hypotheses. These are:**

  (i) $H1$: weight≠100

  (ii) $H1$:weight>100 and

  (iii) $H1$:weight<100.
- **We have to consider one of these three hypotheses in a problem.**

Alternative Hypothesis

- Example-The alternative hypothesis is that the mean weight is not equal to 100 grams.

Decision Errors in Hypothesis Testing

- The two hypotheses are complementary to each other.
- We accept (or reject) the null hypothesis; this is equivalent to rejecting (or accepting) the alternative hypothesis.
- In this process, we make two types of decision errors, namely:
  - Type I error and
  - Type II error

Decision Errors in Hypothesis Testing

- ***Type I Error-***
- When we reject a null hypothesis when it is true.
- The probability of committing a Type I error is called the significance level denoted by α.

Decision Errors in Hypothesis Testing

- ***Type II Error-***
- When we accept a null hypothesis when it is false.
- The probability of committing a Type II error is denoted by β.
- The probability of not committing a Type II error is called the Power of the test (1-β).

Decision Errors in Hypothesis Testing

	Null Hypothesis is true	Null Hypothesis is False
Reject Null Hypothesis	False Positive (Type I error)	Correct Outcome (True Positive)
Accept Null Hypothesis (or fail to Reject)	Correct Outcome (True Negative)	False Negative (Type II error)

Type-1 and Type-2 Error

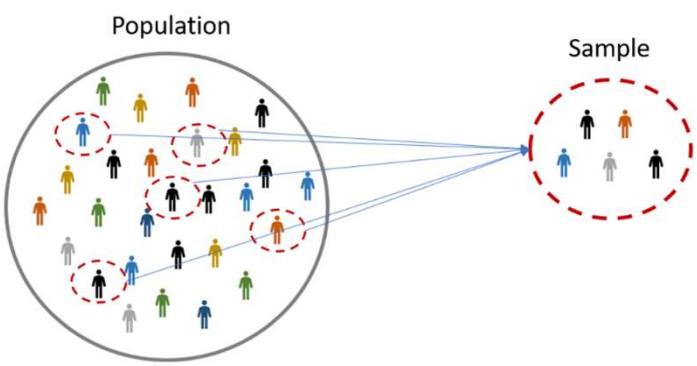
	Null Hypothesis is true	Null Hypothesis is False
Reject Null Hypothesis	False Positive (Type I error)	Correct Outcome (True Positive)
Accept Null Hypothesis (or fail to Reject)	Correct Outcome (True Negative)	False Negative (Type II error)

- <u>Example</u>
- Suppose a teacher evaluates the examination paper to decide whether a student passes or fails
  - H_0 -: Student has passed
  - H_1 -: Student has failed
- Type-I error will be the teacher failing the student [rejects H_0 although the student scored the passing marks [H_0 was true]
- Type-II error will be the case where the teacher passes the student [do not reject H_0 although the student did not score the passing marks [H_1 is true]

Factors Affecting Type I and Type II Error

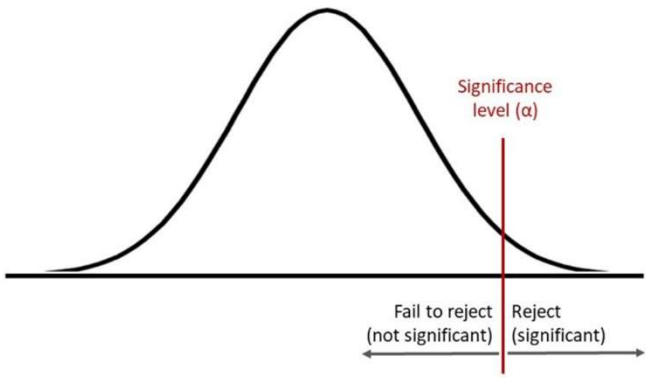
Sample Size

- Larger sample sizes generally reduce the probability of both Type I and Type II errors.
- With larger samples, the estimates tend to be more precise, resulting in more accurate conclusions.



Significance Level (α)

- The significance level (α) in hypothesis testing determines the probability of committing a Type I error.
- Choosing a lower significance level reduces the risk of Type I error but increases the risk of Type II error, and vice versa.
- Most commonly used alpha values are 0.01, 0.05 and 0.1

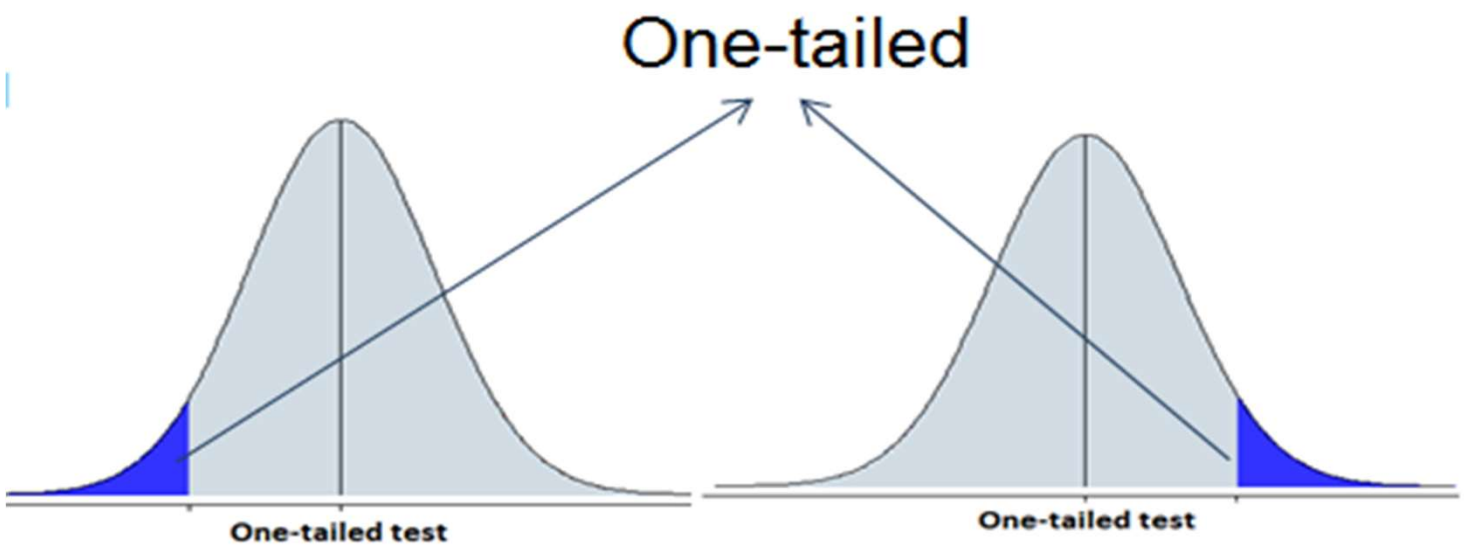


Level of Significance

- The alpha value is a criterion for determining whether a test statistic is statistically significant
- In a statistical test, Alpha represents an acceptable probability of a Type I error.
- Because alpha is a probability, it can be anywhere between 0 and 1
- In practice, the most commonly used alpha values are 0.01, 0.05 and 0.1 which represent a 1%, 5% and 10% chance of a Type I error, respectively (i.e. rejecting the null hypothesis when it is in fact correct)

Hypothesis Testing

- ***One-Tailed Test***
- A test of hypothesis, in which the region of rejection is only on one side of the distribution used for test statistic, is called a one-tailed test.

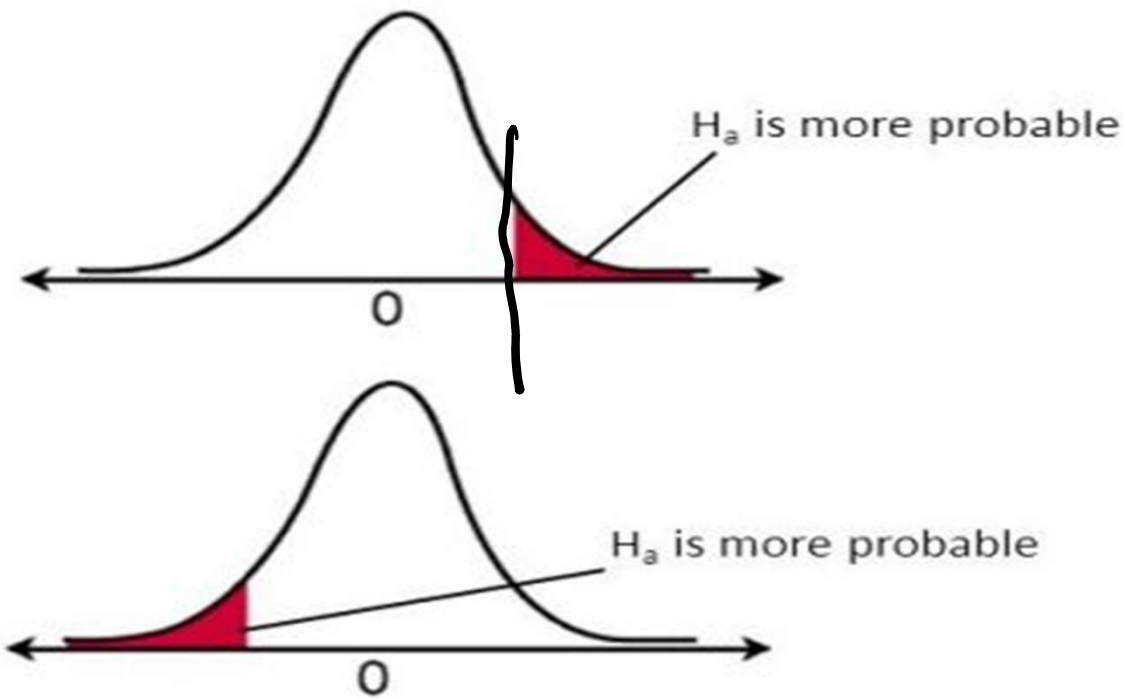


One-Tailed Test

- For example, if we hypothesize that population mean is 12, i.e., null hypothesis is $\mu = 12$.
- We consider that the alternative hypothesis is; mean is less than 12, and then we have to employ a one-tailed test.
- In this situation, alternative hypothesis shall be $\mu < 12$.
- One can see that in some other situation, we may have to consider that $\mu > 12$.

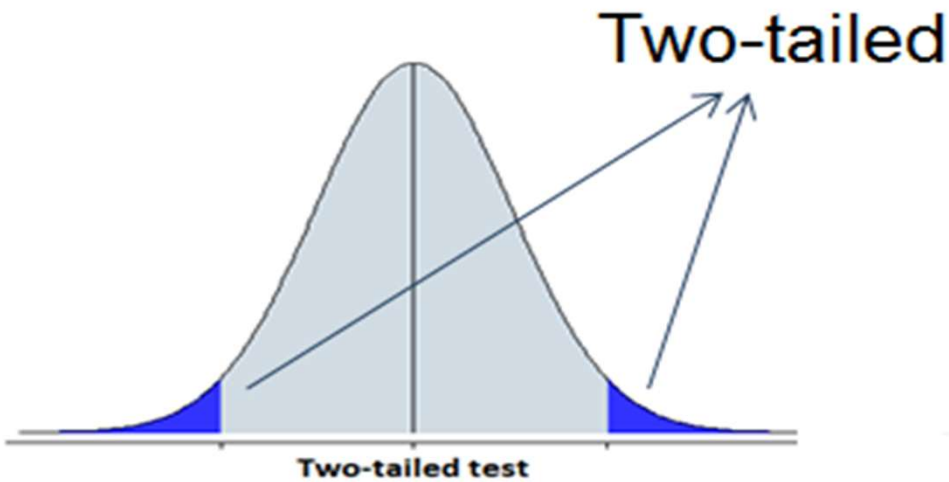
One-Tailed Test

- If the sample being tested falls into the one-sided critical area, the alternative hypothesis will be accepted instead of the null hypothesis.



Hypothesis Testing

- ***Two-Tailed Test***
- A test of hypothesis, in which the region of rejection is on both sides of the distribution used for test statistic, is called a two-tailed test.

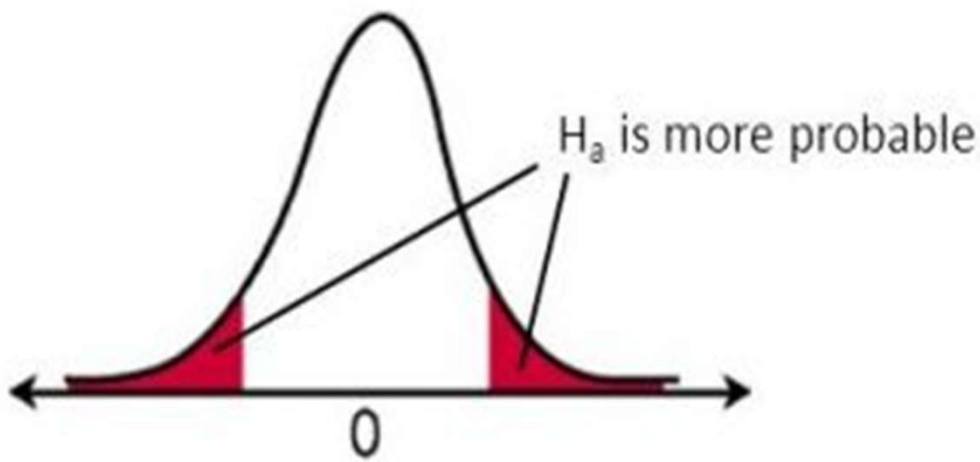


Two-Tailed Test

- For example, if we hypothesize that population mean is 12, i.e., null hypothesis is $\mu = 12$.
- We consider that the alternative hypothesis is; mean is less than 12 or mean is greater than 12, then we have to employ a two-tailed test.
- In such a situation, alternative hypothesis shall be $\mu \neq 12$.

Two-Tailed Test

- If the sample being tested falls into the two-sided critical area, the alternative hypothesis will be accepted instead of the null hypothesis.



Steps in Hypothesis Testing

1. Set up a Hypothesis- State the null and alternative hypotheses.
2. Setup the level of significance- **Significance level is the probability of occurrence of wrong decision. (**If not given then consider 5%.**)**
3. Choose the appropriate test statistic-
   1. Z-Score (sample size is greater than or equal to 30)
   2. t- Test (sample size is less than 30)

Data that approximately follows a normal distribution

Steps in Hypothesis Testing

4. Computation-

Z-Test-

$$Z = \frac{x - np}{\sqrt{npq}}$$

x-observed success, n-sample size, Z-calculated critical value, p-probability of success, q-probability of failure

$$Z = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

$\bar{x}$ is the sample mean, μ is the population mean, σ is the population standard deviation and n is the sample size.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Steps in Hypothesis Testing

4. Computation-
   - t-Test-

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

s- standard deviation of the sample, μ - mean of the population, $\bar{x}$-sample mean

Steps in Hypothesis Testing

5. Making Decision-

Z_cal > Z_tab, hence (H_0) is rejected- For Z-Test

t_cal > t_tab, hence (H_0) is rejected- For t-Test

Note:

- **In two-tailed tests, we compare | t_{cal} | or | Z_{cal} | with t_{tab},or Z_{tab}, respectively not the signed value.**
- **One-tailed test: Decision depends on the sign and magnitude of t_{cal} ,or Z_{cal}, respectively and it is compared with t_{tab}or Z_{tab}, respectivelyin the direction of H_a.**

Z-test vs t-test Assumptions

Z-Test	t-Test
σ known	σ unknown
Large n (≥30)	Small n (<30)

Hypothesis Testing

- A coin was tossed 484 times and Head turned up 265 times. Test the hypothesis that the coin is unbiased. The critical Z-values are given.

p	Significance Level	Confidence Level	Two-*Tailed Test*	One-*Tailed Test*
0.1	10%	90%	1.65	1.28
0.05	5%	95%	1.96	1.64
0.01	1%	99%	2.58	2.33
0.001	0.1 %	99.9%	3.29	3.10

Hypothesis Testing

Solution-

- Step-1 :
  - H_0 – Coin is unbiased, meaning the probability of getting heads is p=0.5
  - H_a – Coin is biased meaning the probability of getting heads is not equal to 0.5 ($p \neq 0.5$).
- Step-2: Setup a significance level- If not given then consider 5%.
- Step-3: Setting a Test statistic- Z-Score (sample size is greater than 30)
- **Z= x-np/sqrt(npq)**

Hypothesis Testing

Solution-

- Step-4: Calculation

  n=484, x=265, p=0.5, q=0.5

  Z= x-np/sqrt(npq)

Put the values:

Z= 265-484.(0.5)/sqrt(484.(0.5).(0.5)

Z_cal =2.09

Z_tab = 1.96

Hypothesis Testing

Solution-

- Step-5: Make a decision
- Check -1: Z_cal > Z_tab, hence (H_0) is rejected.

or

- Check-2: The Z_cal (2.09) does not lie in the range of Z_tab (i.e. -1.96 to 1.96)

Hence the Null Hypothesis (H_0) is rejected.

This means there is sufficient statistical evidence at 5% significance level to conclude that the coin is biased.

Hypothesis Testing

- In 256 throws of a six faced dice, odd points appeared 122 times. Would you say that the dice is fair at 5% level of significance?

Hypothesis Testing

- Solution-

$$Z_cal = -0.75 < Z_tab$$

Hence H_0 is accepted.

This means there is sufficient statistical evidence at 5% significance level to conclude that the dice is fair.

Hypothesis Testing

Q.3.1 A professor claims that all the students in the first-year class possess average IQs. Randomly, a test was conducted on thirty students, resulting in a mean IQ is 117. The population mean was 100, and the standard deviation was 27.

Q.3.2 A professor claims that first-year students have higher-than-average Iq. To test this claim, a random sample of 30 students was taken, and their mean IQ was found to be 117. The population mean IQ is known to be 100, with a standard deviation of 27. Conduct a hypothesis test at a 5% significance level to determine if the students have significantly higher IQs than the population average.

Hypothesis Testing

Q.3.1- A professor claims that all the students in the first-year class possess average IQs. Randomly, a test was conducted on thirty students, resulting in a mean IQ is 117. The population mean was 100, and the standard deviation was 27. **(Two tailed)**

Q.3.2-A professor claims that first-year students have higher-than-average IQs. To test this claim, a random sample of 30 students was taken, and their mean IQ was found to be 117. The population mean IQ is known to be 100, with a standard deviation of 27. Conduct a hypothesis test at a 5% significance level to determine if the students have significantly higher IQs than the population average.

(One tailed)

Hypothesis Testing- Question-Solution 3.1

- H_0 – All the students in the first-year class possess average Iqs (IQ=100)
- H_a – All the students in the first-year class do not possess average Iqs (IQ ≠100)

Setup a significance level- If not given then consider 5%.

$$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{117-100}{\frac{27}{\sqrt{30}}} = \frac{17}{\frac{27}{5.47}} = \frac{92.99}{27} = 3.44$$

Hypothesis Testing

- **Solution 3.1- Two tailed test**

$$Z = \frac{\bar{x}-\mu}{\frac{\sigma}{\sqrt{n}}} = \frac{117-100}{\frac{27}{\sqrt{30}}} = \frac{17}{\frac{27}{5.47}} = \frac{92.99}{27} = 3.44$$

$$\mathbf{Z_{Cal}} = 3.44$$

Z_{Tab} =1.96 (Value from the Table given in question 1)

$Z_{cal} > Z_{tab}$, hence (H_0) is rejected.

This means there is sufficient statistical evidence at 5% significance level to conclude that the first-year class do not possess average IQs.

Hypothesis Testing

- **Solution 3.2**

Step 1: State the hypotheses

The claim is *"higher-than-average IQs"*.

$$H_0: \mu = 100$$
$$H_a: \mu > 100 \ (Right-tailed\ test)$$

Do all the steps from 1 to 5 (in the MST)

Step 4: Decision

$$Z_{cal} > Z_{tab}$$
$$3.45 > 1.645$$

There is sufficient statistical evidence at the **5% level of significance** to conclude that **first-year students have significantly higher IQs than the population average**.

Hypothesis Testing

- The manufacturer of a certain make of Thermometer claims that his thermometers have a mean life of 20 months. A random sample of 7 such thermometers gave the following values. Life of thermometers in months: 19, 21, 25, 16, 17, 14, 21. Can you regard the producer's claim to be valid at 1% level of significance? Given that t_tab= 3.707.

Solution- (Two Tailed Test)

19, 21, 25, 16, 17, 14, 21

- Mean of the sample = $\frac{19 + 21 + 25 + 16 + 17 + 14 + 21}{7}$ = 19

- Standard Deviation (s) = $\sqrt{\frac{\sum(x_i - \mu)^2}{N-1}}$

- s = 3.69
- Step-1
  - H_0= Thermometers have a mean life of 20 months. $(\mu = 20)$
  - H_a= Thermometers don not have a mean life of 20 months. $(\mu \neq 20)$

Hypothesis Testing-Solution

- μ = 20, n=7, calculate $\bar{x}$ which is the mean of the sample using the data given in the question.
- Then put all the values in the below equation:

- $$t = \frac{\bar{x}-\mu}{\frac{s}{\sqrt{n}}} = \frac{19-20}{\frac{3.69}{\sqrt{7}}} = \frac{-1}{\frac{3.69}{2.64}} = \frac{-2.64}{3.69} = -0.71$$

- t_{cal} = -0.7 and t_{tab}= 3.707
- H_0 is accepted ($|t_{cal}| < t_{tab}$)

This means there is sufficient statistical evidence at 1% significance level to conclude that thermometers have a mean life of 20 months.